

1. What were the earliest known methods of cryptography used in ancient civilizations?
 - Ancient people came up with clever ways to hide their messages from enemies and competitors. The Spartans created the scytale around 400 BCE - they'd wrap a strip of leather around a wooden rod, write their message, then unwrap it so the letters looked jumbled unless you had a rod of the exact same size. Egyptians changed their hieroglyphs in tomb writings to make things look mysterious, while Hebrew scholars made the Atbash cipher that basically reversed the alphabet. Mesopotamians even hid their recipes for making pottery glazes to keep their trade secrets safe, showing that people have always needed ways to protect important information.
2. How did the Caesar cipher work, and why was it considered secure at the time?
 - Julius Caesar came up with a simple way to hide his military messages by moving each letter forward in the alphabet by a certain number of positions - he might shift by 2 or 3 spots, whatever he decided. If he picked 2, then A becomes C, B becomes D, C becomes E, and at the end Y becomes A and Z becomes B. This actually worked pretty well back then because hardly anyone in ancient Rome could even read, much less figure out coded messages. Your enemies would need to find someone who understood Latin and knew about secret codes, which was super rare, so Caesar's method kept his military plans safe.
3. What role did cryptography play during major historical events such as World War II?
 - Cryptography played a huge role in World War II and actually helped decide who won. The Germans used the Enigma machine to send secret messages and thought nobody could crack it, but Alan Turing and his team at Bletchley Park managed to break the code and read German military plans. Getting access to these secret messages helped the Allies avoid German submarines and plan major operations like D-Day, and historians think it might have ended the war 2-4 years earlier. The Americans also broke Japan's secret codes, which gave them information that helped them win important battles like Midway.
4. How did the invention of the Enigma machine influence modern cryptography?
 - The Enigma machine changed everything by showing that machines could create complicated codes - it had rotating wheels that changed how each letter was scrambled every time you typed. Breaking these codes pushed scientists to build some of the first real computers, like the Colossus machines at Bletchley Park. The work of cracking Enigma taught us important lessons we still use today, like changing your codes regularly and that smart mathematical thinking can break even complicated systems. After the war, governments realized that good cryptography was essential for protecting their countries, so they started spending a lot more money developing better encryption.
5. What were the limitations of classical cryptographic techniques before the digital era?
 - Old encryption methods had serious problems that made them easy to break. Once someone figured out frequency analysis in the 9th century, simple codes became easier to crack because some letters show up way more than others in any language. There was also the problem of getting the secret key to the person you wanted to talk to - if you could safely meet them to hand over the key, why not just tell them the message directly? People had to do all the encryption by hand, so the codes couldn't be too complicated, or they'd take forever and have mistakes, which meant patient codebreakers could eventually figure them out. These old methods also had no good way to prove who sent a message.

6. How did the development of computers change the landscape of cryptography?
 - Computers totally changed cryptography because they can do millions of math problems in seconds, which means you can use complicated math that humans could never do by hand. This made new encryption methods like DES and AES possible - they use complex math operations that only computers can handle fast enough to be practical. In the 1970s, someone invented public key cryptography, which finally solved the problem of sharing keys safely by using hard math with giant prime numbers. Computers also made it easier to crack weak codes quickly, so encryption experts had to keep making stronger systems to stay ahead. Everything digital today - your bank account, online shopping, text messages - relies on computer-powered encryption to stay private.
7. What are the key differences between symmetric and asymmetric cryptography, and when is each used?
 - Symmetric encryption is like a regular lock where the same key both locks and unlocks the door - it's fast and works great, but you need to safely give that key to anyone who needs to unlock it. Asymmetric encryption uses two different keys - one public key that anyone can have and one private key that you keep secret, which fixes the key-sharing problem but takes a lot more computer power and time. You use symmetric encryption for protecting big files or keeping a connection secure once it's started, while asymmetric encryption is for starting those secure connections, proving who you are, or safely sharing the symmetric key at the beginning. Most systems today use both - your browser uses asymmetric encryption to safely swap a symmetric key with a website, then switches to symmetric encryption for everything else because it's way faster.
8. How do hash functions ensure data integrity, and what properties make them secure?
 - Hash functions convert any data into a unique fixed-length code, and changing even the smallest detail in the original data creates a completely different code, making tampering easy to detect. For security, hash functions must be impossible to reverse, extremely difficult to find two different inputs that create the same hash (collision-resistant) and always produce the same result for identical inputs. Popular examples like SHA-256 are used for verifying downloaded files and protecting stored passwords. Security comes from mathematical complexity that would require astronomical computing power to break.
9. What is a digital signature, and how does it provide authenticity and non-repudiation in communication?
 - A digital signature works by creating a fingerprint of your message and encrypting it with your private key, which recipients can verify using your public key to confirm the message is authentic and unchanged. This proves your identity as the sender because only you possess your private key, and it provides non-repudiation, meaning you can't deny sending the message later since the signature is mathematical proof. Digital signatures are used for electronically signing contracts, legal documents, and verifying software hasn't been tampered with. The entire process ensures both authenticity and integrity of communications.
10. How does Public Key Infrastructure (PKI) establish trust between parties, and what are its main components?

- PKI is a trust system where Certificate Authorities (CAs) verify identities and issue digital certificates that bind public keys to verified identities, while Registration Authorities handle the verification process. When you visit a secure website, your browser checks the site's certificate to confirm it was issued by a trusted CA, hasn't expired, and belongs to the legitimate owner. This creates a chain of trust where you trust root CAs, who trust intermediate CAs, who then vouch for end-user certificates. The system makes secure internet communication possible by letting you verify that websites and people are really who they claim to be.
11. What are the common challenges and security risks associated with implementing these cryptographic concepts in real-world systems?
- Implementing cryptography is difficult because people often mishandle encryption keys by losing them, storing them insecurely, or using weak passwords, which undermines even the strongest encryption. Developers frequently make implementation mistakes like using outdated algorithms, not validating certificates properly, or creating security holes, while users pick weak passwords, reuse them, or fall for phishing attacks. Attackers exploit side-channel vulnerabilities by measuring timing differences or electromagnetic emissions rather than breaking the actual math. Organizations must also balance strong security with system performance and user experience, which often leads to compromises.
12. How are asymmetric keys managed throughout their lifecycle, and what challenges and security risks are involved?
- Key management involves generation (requiring truly random numbers), secure storage and distribution of private keys while making public keys verifiable, protection against theft through malware or social engineering, and periodic renewal to limit damage from eventual compromise. When keys are compromised, revocation systems must notify everyone to stop trusting them, but these warnings don't always reach people quickly enough. Organizations often manage thousands or millions of keys, and losing track of even one can create serious security vulnerabilities. The biggest challenge is balancing security, accessibility, and backup needs throughout each key's entire lifespan.
13. What are the major cryptographic threats and emerging issues—such as quantum computing—and how are they shaping the future of cryptography?
- Quantum computing is the biggest threat because it could break most current encryption methods, and attackers are already using "harvest now, decrypt later" strategies to steal encrypted data today and wait for quantum computers to crack it. Researchers are developing post-quantum cryptography and pushing transition systems before quantum computers become powerful enough, while also addressing attacks on AI systems and weak security in IoT devices. There's a critical need for crypto-agility—building systems that can quickly switch to new algorithms when current ones are broken. These threats are driving development of more resilient algorithms and forcing organizations to plan for long-term security rather than just immediate needs.